Toward Cross-Domain Automated Feedback: A Comparative Evaluation of Open-Source Models Across Diverse Student Assessment Types

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One open-source model cannot do everything — the best model depends on the domain, and multi-agent prompting is not always better

Background & Motivation

The Problem

Most LLM-based feedback tools rely on proprietary models (GPT-4, Claude, Gemini), limiting transparency, reproducibility, and access in resource-constrained settings.

The Gap

Most existing systems are designed for a single educational domain, making them impractical as unified classroom tools and leaving cross-domain generalizability largely unexplored.

Our Approach

We evaluate two open-source models under two prompting strategies across three educational domains using a unified human annotation framework

INPUT

Educational Datasets

Programming
ConDefects
n=200
Python (93) | Java (107)
Buggy code + Ground truth

Writing
Exposia
n=55
Bachelor thesis proposals
26 criteria across 4 dimensions

Mathematics
MathEDU
n=150
Word problems
Student steps + Teacher review

Data Preparation
Problem + Submission + Criteria

Implementation Framework

Models & Prompting Approaches

LLM MODELS (~30B parameters)

Qwen 3.5 27B | DeepSeek R1 32B

Unified Approach: All-in-One Processing → Final Feedback

Multi Agent Approach: Diagnoser → Feedback Gen → Final Feedback

Task Decomposition: Broken into sub-problems.

Functional Experts: (e.g., 2-5 specialized roles).

EVALUATION

Human Annotation Framework

ANNOTATORS
Graduate students with domain expertise

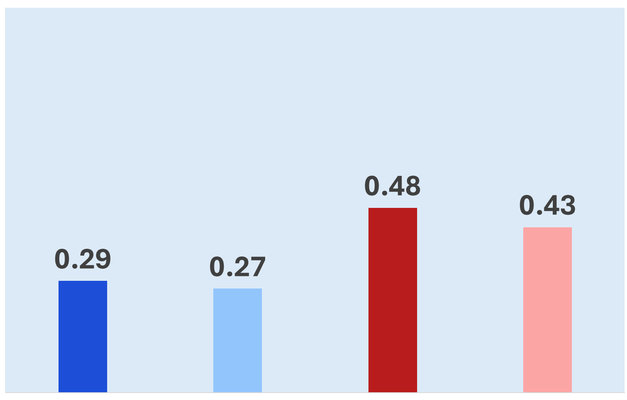
6 Criteria per Domain (Binary 0/1)

DIAGNOSIS (Error Identification)
Accuracy - Correct issue identified
Selectivity - No false positives
Clarity - Pedagogically accessible

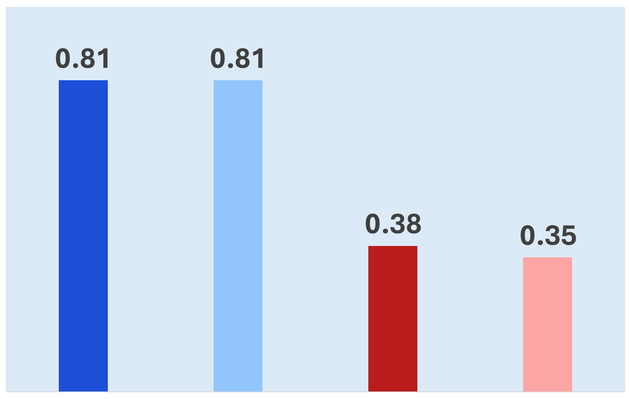
GUIDANCE (Corrective Feedback)
Accuracy - Valid correction provided
Selectivity - Focused, specific advice
Clarity - Actionable guidance

Results & Key Findings

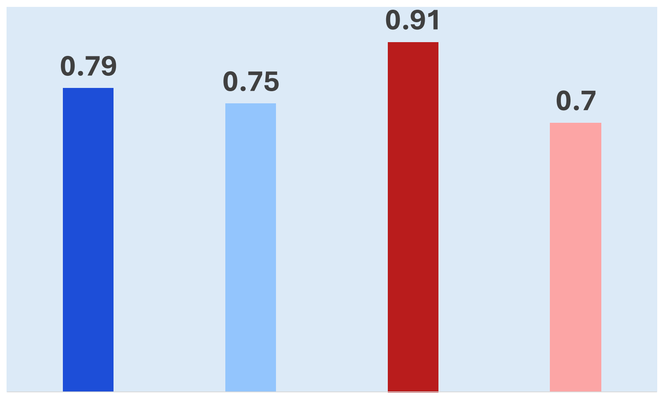
Qwen – U = Qwen 3.5 with Unified prompting | Qwen - MA = Qwen 3.5 with Multi-Agent | DS – U = Deepseek R1 with Unified prompting | DS – MA = Deepseek R1 with Multi-Agent



Programming (ConDefects, n=200)



Writing (Exposia, n=55)



Mathematics (MathEDU, n=150)

Programming Insights
EC high (0.82–1.00), EA low (0.12–0.20)
Fluent explanations ≠ correct diagnosis.

Writing Insights
WA/GA high for Qwen (≈0.9)
Accurate issue detection drives feedback quality.

Mathematics Insights
MA ≈ 1.0 for both models
Performance differs in solution guidance.

Multi Agent Impact:
Programming: Neutral
Writing: Neutral
Mathematics: Reduced Solution Guidance (SA)
Task decomposition was not universally beneficial.

Qualitative Analysis: Feedback on Competitive Programming Solutions

K-th largest in subtree (K ≤ 20 — only top-20 per node needed)

```
for i in S[to_x]:
    if len(S[x]) < 20:
        S[x].add(i)
    else:
        m = min(S[to_x]) ← BUG: should be min(S[x])
        if i > m:
            S[x].discard(m)
            S[x].add(i)
```

Unified Approach

Your SortedMultiset approach is on the right track.

Bug identified: In the merge loop, you compare each child element against the **current minimum of the parent**, potentially discarding valid child values. The parent set may miss entries that should replace its smallest element.

Conceptual issue: Inserting values one-by-one with a conditional does not guarantee the parent retains the true top-K.

Fix: Merge all child elements into the parent, then remove the minimum until size = 20.

✗ *Misidentifies source: code uses `min(S[to_x])` — the child's min, not the parent's.*

Multi-Agent Approach

Good use of SortedMultiset to track top-K candidates per node.

Bug identified: `m = min(S[to_x])` uses the child's own minimum, not the parent's. Child values > child-min but < parent-min are incorrectly excluded from the parent's top-20.

Conceptual issue: The top-K invariant is violated because the threshold references the wrong set.

Fix: Add all child elements to the parent set, then remove minimums until size = 20.

✓ *Correctly identifies `min(S[to_x])` as the child's minimum.*

EA	Error Accuracy
EC	Explanation Clarity
WA	Writing Accuracy
GA	Guidance Accuracy
MA	Mathematical Accuracy
SA	Solution Accuracy

Conclusion & Future Work

Conclusion

- Model selection matters more than prompting strategy.
- **DeepSeek R1** excels at structured tasks (math: 0.91, programming: 0.48)
- **Qwen 3.5** performs best on open-ended tasks (writing: 0.81)
- Models explain bugs clearly (Clarity=1.00) **yet rarely identify them correctly** (Accuracy=0.12–0.20) — students receive **confident but misleading feedback.**
- **LLM-as-judge guardrails** can automatically filter low-quality feedback before it reaches students.

Limitations

- Offline evaluation only — no classroom deployment yet, so actual impact on student learning outcomes remains unverified.
- Only 2 open-source models evaluated across 3 relatively small datasets, which limits generalizability to larger models and broader contexts
- Binary annotation scheme may miss partial correctness and nuanced quality differences in generated feedback.
- Annotators had domain expertise but no pedagogical training, and inter-annotator agreement was calculated on limited overlapping samples only.

Future Work

- **Classroom deployment** to measure direct impact on student learning outcomes.
- **Deploy automated feedback evaluation** as scalable quality control alongside human oversight.

Live Demo

Code & Data

- huggingface.co/spaces/mhaseeb/Academic_Feedback_Generator
- github.com/haseeb2001/crossdomainfeedback